MIL-P-22438B(SH)
10 May 1977
SUPERSEDING
MIL-P-22438A(SHIPS)
10 July 1967
(See 6.5)

MILITARY SPECIFICATION

PANELS, RECEPTACLE, FOR ELECTRICAL AND ELECTRONICS

TEST AND MAINTENANCE TABLE, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for electrical and electronic test and maintenance table receptacle panels.

1.2 Classification. Panels shall be of the types specified in the individual specification sheets (see 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-C-596 - Connector, Plug, Electrical: Connector, Receptacle, Electrical.
W-C-596/12 - Connector, Receptacle, Electrical, General Purpose, Duplex Grounding, 2 Pole, 2 Wire, 15 Amperes, 125 Volts.
UU-P-268 - Paper, Kraft, Wrapping Cushioning Material.
PPP-C-850 - Cushioning Material, Polystyrene Expanded, Resilient (For Packaging Uses).
PPP-C-1120 - Cushioning Material, Uncompressed Bound Fiber For Packaging.

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements For (Naval Shipboard Use).
MIL-R-2726 - Receptacles, Receptacle Plugs, Switch and Receptacles, and Outlets (Electrical), General Specification For.
MIL-C-5015 - Connectors, Electrical, "AN" Type, General Specification For.
MIL-R-6130 - Rubber, Cellular, Chemically Blown.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-E-15090 - Enamel, Equipment, Light Gray (Formula No. 111).
MIL-P-15328 - Primer, (Wash) Pretreatment Blue (Formula No. 117-B for Metals).
MIL-F-16377 - Fixtures, Lighting; and Associated Parts Shipboard Use, General Specification for.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of.
MIL-R-20092 - Rubber Sheets and Molded Shapes, Cellular, Synthetic Open Cell (Foamed Latex).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6110

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
PB 1 - Panelboards.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish sample panel for first article inspection and approval (see 4.4 and 6.3).

3.3 Material. The material for each part shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the receptacle panels to meet the performance requirements of this specification.

3.3.1 Restricted material. Flammable or explosive material, or material which can produce toxic or suffocating fumes shall not be used. Mercury in any form shall not be used in shipboard equipment, including materials and parts.

3.3.2 Metals. Current-carrying parts and terminal posts shall be copper alloy. Other metal posts shall be either noncorrosive or shall be treated, coated, or plated in accordance with MIL-E-917 to prevent corrosion.

3.3.2.1 Dissimilar metals. Dissimilar metals in contact with each other shall be in accordance with the requirements of MIL-E-917.

3.3.3 Terminal board insulating materials.

3.3.3.1 Plastics, laminated. Laminated plastic material shall be glass base laminated plastic in accordance with MIL-E-917.

3.3.3.2 Plastics, molded. Molded terminal boards shall be made of glass fiber reinforced thermosetting molding compounds in accordance with MIL-E-917.

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3.3.3.3 Flash shall be removed from molded parts and the width of the flash edges made thereby shall be not greater than 1/32 inch. Flash edges shall be buffed smooth.

3.4 Design and construction. Receptacle panels shall conform to the design, construction, and physical dimensions specified on the specification sheets.

3.4.1 Hardware. Screws, nuts, washers, and miscellaneous hardware shall be of a good commercial grade material compatible with that of the basic panel, and protected against corrosion. Self-tapping screws or other self-tapping devices shall not be used.

3.4.1.1 Locking devices. Locking devices shall be used, where required, to secure components to insure continuous satisfactory operation under the test conditions specified.

3.4.2 Threads. Threads shall be right hand, class 2, complying with Unified and American national screw threads as specified in Handbook H28.

3.4.2.1 Engagement of threaded parts. Thread engagement shall be a minimum of 2/3 of diameter of bolt or screw.

3.4.3 Drilling, countersinking, and tapping. All drilling, countersinking, and tapping shall be done before plating or finish is applied, except when necessary to insure conformance with 3.4.5.4.

3.4.4 Sharp edges. All sharp edges that are accessible shall be given a slight radius.

3.4.5 Receptacle and plug assemblies.

3.4.5.1 Receptacles direct current (d.c.) and 400 hertz (Hz) alternating current (a.c.). Receptacles furnished for d.c. and 400 Hz power outlets shall be in accordance with MIL-C-5015 or MIL-R-2726, as applicable. The type receptacle designated by part number or specification sheet number, and the quantity to be furnished shall be as specified on the applicable specification sheet.

3.4.5.2 Receptacles, 60 Hz a.c.. Receptacles for 60 Hz power outlets shall be duplex receptacle assemblies in accordance with W-C-596 and W-C-596/12 and furnished in the quantity specified on the applicable specification sheet.

3.4.5.3 Plug assemblies, d.c. and 400 Hz a.c.. Plug assemblies for use with d.c. and 400 Hz outlets shall be furnished only when specified on the individual specification sheet. The plugs, when specified, shall be in accordance with MIL-C-5015 or MIL-R-2726 and furnished in the quantity specified on the applicable specification sheet.

3.4.5.4 Ground continuity. Any receptacle with a ground contact electrically connected to the receptacle shell or mounting strap shall maintain this electrical continuity with the panel through the mounting screws (an additional connection to a ground post may be required by the specification sheet).

3.4.6 Terminal boards. Terminal post assemblies (see 3.4.6.1) shall be securely fastened to or molded integrally with the terminal board base by a method that will prevent any loosening or change in position under the test conditions of 4.7.5 and 4.7.6. Any live part of the terminal board exposed on the under side of the base (such as screws, rivets, and so forth used for attaching terminal post assemblies to the base) shall be recessed in a counterbored hole, and covered with a good-quality insulating resin. The sealing of the counterbored holes with resin is not required when the fasteners involved are prevented from loosening under vibration shock, and an insulated subbase is employed, or the creepage and clearance distances specified in MIL-E-917 are adequate without the use of a subbase. To facilitate interpanel bus connections, the orientation of bus connection terminal posts shall be as shown on the applicable specification sheet. Phase and frequency, or polarity markings, as shown on the specification sheet, shall be molded or engraved on the terminal boards. The height of the terminal boards, after installation of the panel, shall be sufficient to permit interpanel bus wiring to clear receptacles, if the terminal board location and terminal post orientation requires this wiring to pass over the receptacles (see applicable specification sheet).

3.4.6.1 Terminal posts. Separate terminal connectors or posts, silver brazed (except in the case of screw terminal connectors) to a common terminal bus bar, shall be provided for interpanel bus connections and local interior wiring. The terminal posts used for interpanel wiring shall be as specified on the applicable specification sheet. The type of terminal connector used for interior wiring shall be at the option of the manufacturer.

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3.4.6.1.1 Ground connection. A separate ground connection terminal post shall be brazed to the panel or panel enclosure in the location and orientation shown on the specification sheet.

3.4.7 Interior wiring. All interior wiring shall be furnished and installed by the manufacturer using stranded copper wire of the type required by the individual specification sheet. Connections at screw terminals shall be made with pressure type connectors. Connectors shall be prevented from turning, unless such turning does not decrease the creepage and clearance distances below those specified in MIL-E-917. Interior wiring shall be connected as shown in the wiring diagram on the applicable specification sheet.

3.4.8 Wiring diagram. A wiring diagram shall be furnished with all panels, and attached to the rear of the panel or to the inside of the cover for a totally enclosed panel. A type III label, as specified in MIL-P-16377, shall be used.

3.4.9 Enclosed receptacle panels. Receptacle panels designated as enclosed shall have all their components housed in a type 1 general purpose steel enclosure in accordance with NEMA PB 1, and finished in accordance with 3.7. Enclosure cover or door shall be sized to permit easy access to all interior components. A locking device shall be provided if the method of attachment of the cover or door to the enclosure is by hinges or means other than screw type fasteners. The manufacturer is cautioned that the thickness of the sheet metal and the supporting channels or strips used in the enclosure must be sufficient to permit the enclosure, complete with all interior components, to comply with the test conditions of 4.7.5 and 4.7.6.

3.5 Performance.

3.5.1 Operation. Receptacle panels shall perform in satisfactory manner when tested as specified in 4.7.1.

3.5.2 Dielectric strength. When receptacle panels are tested as specified in 4.7.2 there shall be no arcing, breakdown, or other evidence of damage or malfunction.

3.5.3 Insulation resistance. When receptacle panels are tested as specified in 4.7.3 the insulation resistance between electrical conductors and between electrical conductors and ground shall be not less than 100 megohms.

3.5.4 Temperature rise. When enclosed receptacle panels are tested as specified in 4.7.4 the temperature rise of any current carrying component shall not exceed 30°C.

3.5.5 Vibration. When receptacle panels are tested as specified in 4.7.5 there shall be no evidence of cracking, breaking, loosening of parts, interruption of circuit continuity, or other damage.

3.5.6 Shock. When receptacle panels are tested as specified in 4.7.6 there shall be no loosening of parts or other damage and the panels shall be capable of performing their intended function.

3.6 Identification plates and labels. All open receptacle panels shall have an identification label, type I as specified in MIL-P-16377, attached to the interior of the panel. Enclosed receptacle panels shall be provided with either two of these identification labels, attached to the interior and exterior of the enclosure cover, or one identification plate, in accordance with MIL-P-15024 and MIL-P-15024/5, types A, B, C, and F only, mounted externally on the enclosure cover. The identification plate or label shall contain the following information:

(a) Name of the item.
(b) Complete electrical rating.
(c) Specification sheet number and complete Military part number as indicated on the appropriate specification sheet.
(d) National stock number.
(e) Contract number.
(f) Manufacturer's (prime contractor) name or monogram.

3.7 Finish. The main panel or enclosure shall be given one coat of wash primer in accordance with MIL-P-15328, followed by two coats of gray equipment enamel conforming to MIL-E-15090, type II or, type III, class 2.

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8. Amend the drawing as follows:

3.8 Panel marking. White lettering shall be applied by engraving, stenciling or screening (see specification sheet for location and text) directly on the panel, or separate information plates, in accordance with MIL-P-15024 and MIL-P-15024/5 types A, B, C, F, or H, and may be secured to the panel or enclosure. Letters shall not be less than 3/32 inch high.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contrator is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.2 Inspection conditions. Unless otherwise specified, (see 6.2.1), inspection shall be performed in accordance with the test conditions of MIL-STD-202.

4.3 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.4 First article inspection. First article inspection shall subject one panel to the examination and tests specified in table I in the order shown.

TABLE I. First article inspection.

Inspection	Requirement paragraph	Examination and test paragraph
Visual and mechanical examination	3.1, 3.3, 3.4, 3.6, 3.7 and 3.8	4.6
Operation	3.5.1	4.7.1
Dielectric strength	3.5.2	4.7.2
Insulation resistance	3.5.3	4.7.3
Temperature rise	3.5.4	4.7.4
Vibration	3.5.5	4.7.5
Shock	3.5.6	4.7.6

4.4.1 First article test reports. Test reports shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in table II, in the order shown.

TABLE II. Quality conformance inspection.

Inspection	Requirement paragraph	Examination and test paragraph
Visual and mechanical examination	3.1, 3.3, 3.4, 3.6, 3.7, and 3.8	4.6
Operation	3.5.1	4.7.1
Dielectric strength	3.5.2	4.7.2
Insulation resistance	3.5.3	4.7.3

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4.5.1 Lot. A lot shall consist of all panels covered by a single specification sheet produced under essentially the same conditions, and offered for inspection at one time.

4.5.2 Sampling. Sampling shall be in accordance with inspection level II of MIL-STD-105. The acceptable quality level (AQL) shall be 1.5 and 4.0 percent defective for major and minor defects respectively. Major and minor defects shall be as defined in MIL-STD-105.

4.6 Examination. The receptacle panel shall be subjected to a visual and mechanical surface examination to ascertain that the requirements of this specification and the applicable specification sheets concerning materials, design, construction, dimensions, and marking, have been met.

4.7 Tests.

4.7.1 Operation. Each receptacle panel shall be coupled with its appropriate plugs and inspected for proper electrical and mechanical operation.

4.7.2 Dielectric strength. An a.c. root mean square (rms) test voltage of 60 hertz frequency, equal to 1000 volts plus twice the rated voltage shall be applied between points of opposite polarity and also between live parts and ground. The test voltage shall be increased from zero at a rate not exceeding 500 volts per second until the specified potential is reached and shall be maintained at that value for one minute. The source of test potential shall have a rating of at least one kilowatt.

4.7.3 Insulation resistance. Insulation resistance of receptacle panels shall be measured with a 500 volt d.c. megometer, between electrical conductors and between electrical conductors and ground, at an approximate room temperature of 25°C. and at relative humidity of approximately 50 percent.

4.7.4 Temperature rise. Each current carrying component shall carry its maximum rated current, with no parts showing a temperature rise in excess of that specified in 3.5.4. The test current shall be applied until a stable temperature is obtained; temperature is considered stable when three successive readings, taken at 5-minute intervals, show no change. The temperature rise is the difference between the stable temperature and the average ambient temperature during the last quarter of the test. Temperature readings may be obtained by thermocouples or by other standard methods. Specified components and wiring shall be excluded from this test.

4.7.5 Vibration. Receptacle panels shall be tested in accordance with MIL-STD-167-1, type 1.

4.7.6 Shock. Receptacle panels shall be tested in accordance with MIL-S-901, class 1, type A.

4.8 Inspection of preparation for delivery. Sample packages and packs and the inspection of packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation and packaging, packing and marking. The equipment shall be preserved-packaged level A or C, packed level A, B, or C as specified (see 6.2.1) and marked in accordance with MIL-E-17555.

5.2 Cushioning, dunnage and wrapping materials.

5.2.1 Level A preservation-packaging, level A and B packing. Use of all types of loose-fill materials for applications such as cushioning, filler, stuffing and dunnage for material destined for shipboard use is prohibited.

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5.2.2 Level C preservation-packaging and packing. Unless otherwise specified in the contract or order (see 6.2.1), use of all types of loose-fill materials for applications such as cushioning, filler, stuffing and dunnage for material destined for shipboard use is prohibited. When specified for use by the contract or order, unit packages and containers (interior and exterior) shall be marked or labeled as follows:

"CAUTION

Contents cushioned etc. with loose-fill material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard, or transparent flexible cellular material."

6. NOTES

6.1 Intended use. Receptacle panels covered by this specification are for use on electrical and electronic test and maintenance tables.

6.2 Ordering data.

6.2.1 Procurement documents. Procurement documents should specify the following:

(a) Title, number, and date of this specification and specification sheet, as applicable (see 1.2).
(b) When a first article is required for inspection and approval (see 3.2, 4.2 and 6.3).
(c) Under what conditions inspections shall be performed (see 4.2).
(d) Levels of preservation-packaging and packing required (see 5.1).
(e) When "Caution Label" is used for loose-fill material.

6.2.2 Contract data requirements. When this specification is used in a procurement which incorporated a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following:

Paragraph	Data Requirement	Applicable DID	Options
4.2.1	Inspection system plan	DI-R-4803	--
4.4.1	Report, First article test	UDI-T-23790	--

(Copies of data iter descriptions required by the contractors in connection with specific procurement functions should be obtained from procuring activity or as directed by the contracting officer.)

6.3 First article inspection.

6.3.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3.2 When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a first production item. The first article should consist of one panel. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test, and approval of the first article.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment an lose their separate identity when the equipment is shipped.

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6.5 Changes from previous issue. The symbol "*" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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SUBMITTED BY *(Printed or typed name and address - Optional)*

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